Timothy Choi

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Education

Technical University of Munich — **Munich, Germany**
Exchange — *2024 – 2025*

The University of Adelaide — **Adelaide, Australia**
Bachelor of Mathematical and Computer Sciences, GPA: 6.33/7.0 — *2022 – 2025*

Experience

Dassault Systèmes — **Munich, Germany**
Software Engineer Working Student — *2024 May –*

- Improve the interface of debug tools, reducing development time for the team (C++)
- Streamline the process of test case generation (Python)
- Writing documentation for a code base with ~1 million lines (C++)

Altacomm Technologies — **Adelaide, Australia**
Software Engineer Intern — *2024 Jan – 2024 Mar*

- Empower student communities by streamlining the logistical processes
- Build a full-stack contact us and media release page (Rust, Actix, Astro)

Global Technology Integrator — **Hong Kong**
System Engineer — *2021 Aug – 2021 Nov*

- Lead a team of 4+ people, spearheading the automation and parallelization of our tasks (Batch, Windows AD)
- Help resolve our staff shortage by reducing the manpower and time required by ~200%
- Mentor new staff, helping them understand the technology

Projects

Block Model Compression Algorithm (`tinnamchoi.github.io/projects/bmca`)
C++, Algorithm Design, OOP, Git, Agile

- Implementing octrees to compress 3D data in $O(N \log_8 N)$ time
- Employing agile methodologies for a group of 7 and enforcing good Git practices such as using conventional commits

Binary Search on Color (`tinnamchoi.github.io/projects/bsoc`)
Algorithm Design, JavaScript, HTML, CSS, Web Development

- Using the binary search algorithm to find the color that is closest to the user's desired color
- Combat indecisiveness when it comes to choosing colors

Hare-Clark C++ (`tinnamchoi.github.io/projects/hcc`)
C++, CMake

- Seeing how the CS Club committee was struggling to debug the `hcvote` Python library, I created this C++ library as a sanity check
- In the process of creating this program, I discovered multiple flaws in the original library, and decided to release this as a better alternative

Malcolm bot (`in-progress`)
API, SQLite, Python

- Access the LeetCode, Codeforces and CLIST API to conveniently fetch corresponding information
- Store database of past solved problems per user to generate graphs and statistics

Volunteering

Adelaide Rover Team (`set.adelaide.edu.au/atcsr/australian-rover-challenge`) — **Adelaide, Australia**
Software Developer — *2023 Jul –*

UNIHACK (`unihack.net`) — **Australia (remote)**
Organizer — *2023 Apr –*

The University of Adelaide Computer Science Club (`csclub.org.au`) — **Adelaide, Australia**
Committee member — *2022 Dec –*

Skills & Achievements

Languages

Proficient: C++ — **Intermediate / Beginner**: Bash, C, CSS, HTML, JS, LaTeX, MATLAB, Python

Human: English (native), Cantonese (native), Mandarin (B2), German (A2)

Prizes

Subject prize: Information and Communication Technology 2021 — *Achiever of the best results amongst peers*

Scholarships

University of Adelaide: 30% Global Citizens Scholarship — *Best possible scholarship for international students*